



CC1104-2 – DATA COMMUNICATIONS

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. Which layer is responsible for process to process delivery in a general network model?
A) session layer
B) data link layer
C) transport layer
D) network layer
2. Which topology has a central node connected to all other nodes?
A) bus
B) ring
C) star
D) mesh
3. What was the name of the first network?
A) ASAPNET
B) ARPANET
C) NSFNET
D) CNNET
4. A signal is measured at two different points. The power is P1 at the first point and P2 at the second point. The dB is 0. This means _____.
A) P2 is zero
B) P2 equals P1
C) P2 is much smaller than P1
D) P2 is much larger than P1
5. QPSK system uses a phase shift of _____.
A) π
B) 2π
C) $\pi/4$
D) $\pi/2$
6. Quadrature amplitude modulation (QAM) is a combination of _____.
A) ASK and PSK
B) ASK and FSK
C) FSK and PSK
D) QPSK and PSK
7. In synchronous TDM, for n signal sources of the same data rate, each frame contains _____ slots.
A) n+1
B) n-1
C) 2^n
D) n
8. The _____ technique uses M different carrier frequencies that are modulated by the source signal. At one moment, the sign modulates one carrier frequency; at the next moment, the signal modulates another carrier frequency.
A) FDM
B) DSSS
C) FHSS
D) TDM
9. _____ is an analog multiplexing technique to combine optical signals.
A) FDM
B) TDM
C) WDM
D) CDM
10. The word _____ refers to the portion of a _____ that carries a transmission.
A) channel; link
B) link; channel
C) line; channel
D) line; link
11. _____ is designed to be used in wireless applications in which stations must be able to share the medium without interception by an eavesdropper and without being subject to jamming from a malicious intruder.
A) Multiplexing
B) Encoding
C) Modulation
D) Spread spectrum

12. The _____ technique expands the bandwidth of a signal by replacing each data bit with n bits.
- A) FDMA
B) DSSS
C) FHSS
D) TDMA
13. In _____ framing, there is no need for defining the boundaries of frames.
- A) standard
B) variable-size
C) normal
D) fixed-size
14. HDLC is an acronym for _____.
- A) high-level data link control
B) high-duplex line communication
C) host double-level circuit
D) half-duplex digital link combination
15. In CDMA multiplexing, multiple channels occupy a _____ transmission channel?
- A) different
B) single
C) multiple
D) transparent
16. The required resources for communication between end systems are reserved for the duration of the session between end systems in _____ method.
- A) packet switching
B) circuit switching
C) line switching
D) message switching
17. All the packets in a message follow the different path in _____.
- A) circuit switching
B) virtual circuit networks
C) message switching networks
D) datagram switching
18. In IEEE 802.11, the access method used in the DCF sublayer is _____.
- A) CSMA/CA
B) CSMA/CD
C) CSMA
D) ALOHA
19. The _____ layer of Ethernet consists of the LLC sublayer and the MAC sublayer.
- A) network
B) physical
C) datalink
D) transport
20. An interconnected collection of piconet is called _____.
- A) scatternet
B) mininet
C) micronet
D) multinet

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List the common characteristics of data communication. Show the layers of OSI network model. Explain the basic functionalities of datalink layer.	8	L*2	1	1,2
	b) Define noise. Summarize the types of noises. A signal with 200 milliwatts power passes through 10 devices, each with an average noise of 2 microwatts. What is the SNR? What is the SNRdB?	8	L2	2	1,2
2.	a) Define protocol. Outline the principles followed in protocol layering. Illustrate the encapsulation and decapsulation in protocol layering in the Internet	8	L2	1	1,2
	b) Define latency. Compute the total delay (latency) for a frame of size 5 million bits that is being sent on a link with 10 routers each having a queuing time of 2 μ s and a processing time of 1 μ s. The length of the link is 2000 Km. The speed of light inside the link is 2×10^8 m/s. The link has a bandwidth of 5 Mbps. Which component of the total delay is dominant? Which one is negligible? Justify.	8	L5	2	1,2
3.	a) With neat diagram explain various types of transmission modes in data communication.	8	L2	1	1,2
	b) Explain QPSK implementation with neat diagram.	8	L2	2	1,2

Unit – II

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|-------|---|---|----|---|-----|
| 4. a) | Explain TDM. Two channels, one with a bit rate of 190 kbps and another with a bit rate of 180 kbps, are to be multiplexed using pulse-stuffing TDM with no synchronization bits. Determine the following: | | | | |
| | i. The size of a frame in bits? | | | | |
| | ii. The frame rate? | | | | |
| | iii. The duration of a frame? | | | | |
| | iv. The data rate? | 8 | L5 | 3 | 1,2 |
| b) | List and describe types of HDLC frame. Draw the frame format and explain the fields | 8 | L2 | 4 | 1,2 |
| 5. a) | Apply CRC to detect errors for the given divisor 10011 and the dataword 1101011011. | | | | |
| | i. Find the codeword corresponding to this sequence. | | | | |
| | ii. If the codeword has error in third bit, what does receiver obtain when it does its error checking? | 8 | L3 | 3 | 1,2 |
| b) | Explain different persistence methods used in CSMA | 8 | L2 | 4 | 1,2 |
| 6. a) | What is spread spectrum? With an example, explain direct sequence spread spectrum. | 8 | L2 | 3 | 1,2 |
| b) | Explain the frame format and transitional phases of PPP protocol. | 8 | L2 | 4 | 1,2 |

Unit – III

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|-------|--|---|----|---|-----|
| 7. a) | Discuss the various goals of Fast Ethernet. Describe auto-negotiation feature in fast ethernet. Describe the MAC sublayer features included in Gigabit Ethernet. | 8 | L3 | 5 | 1,2 |
| b) | Explain the architecture of IEEE 802.11 | 8 | L2 | 5 | 1,2 |
| 8. a) | Distinguish between circuit switching networks and datagram networks with an example and necessary diagrams. | 8 | L4 | 5 | 1,2 |
| b) | Explain the architecture of Bluetooth | 8 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
